AI STUDENT PERFOMANCE MODEL: Data Analysis and Predication of Exam Scores

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COURSE: PROGRAMMING OF ARTIFICIAL INTELLIGENCE

PROBLEM STATEMENT

- Understanding how study habits affect academic performance is a common challenge among students. Many learners are unsure how study time translates into exam results.
- This project addresses that gap by building a simple data-driven system that analyzes student study hours and predicts exam performance using machine learning techniques.

AIM AND OBJECTIVES

The main aim of this project is to develop a simple predictive system for student performance.

Objectives:

- To import and process student data using Python
- To perform basic statistical analysis on the dataset
- To visualize the relationship between study hours and exam scores
- To build a prediction model using machine learning
- To evaluate and interpret model results

TOOLS AND TECHNOLOGIES USED

The following tools were used:

- Python programming language
- Pandas for data loading and manipulation
- NumPy for numerical computations
- Matplotlib for graphical visualization
- Scikit-learn for machine learning modeling
- Visual Studio Code as the development environment

DATASET DESCRIPTION

The dataset used in this project is a simple CSV file containing two variables:

- **hours_studied** → input feature used for prediction
- **exam_score** → output variable representing student performance

The dataset was stored in CSV format to ensure compatibility with Python-based data analysis tools.

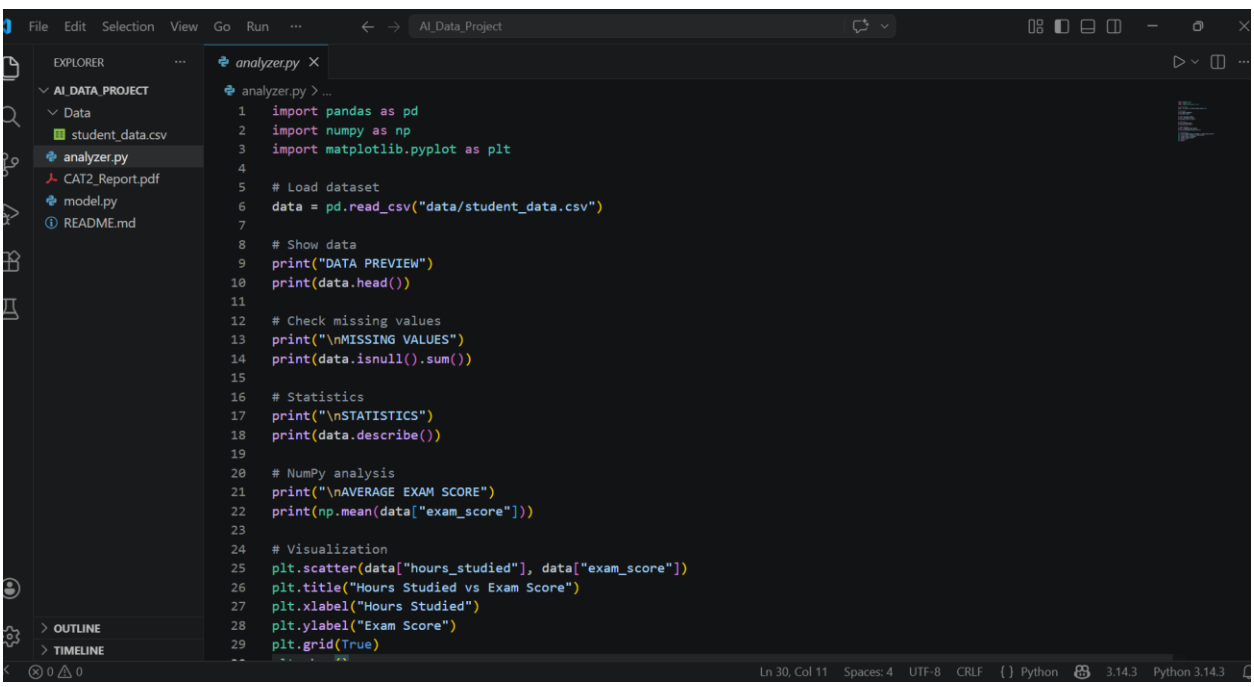
SYSTEM IMPLEMENTATION

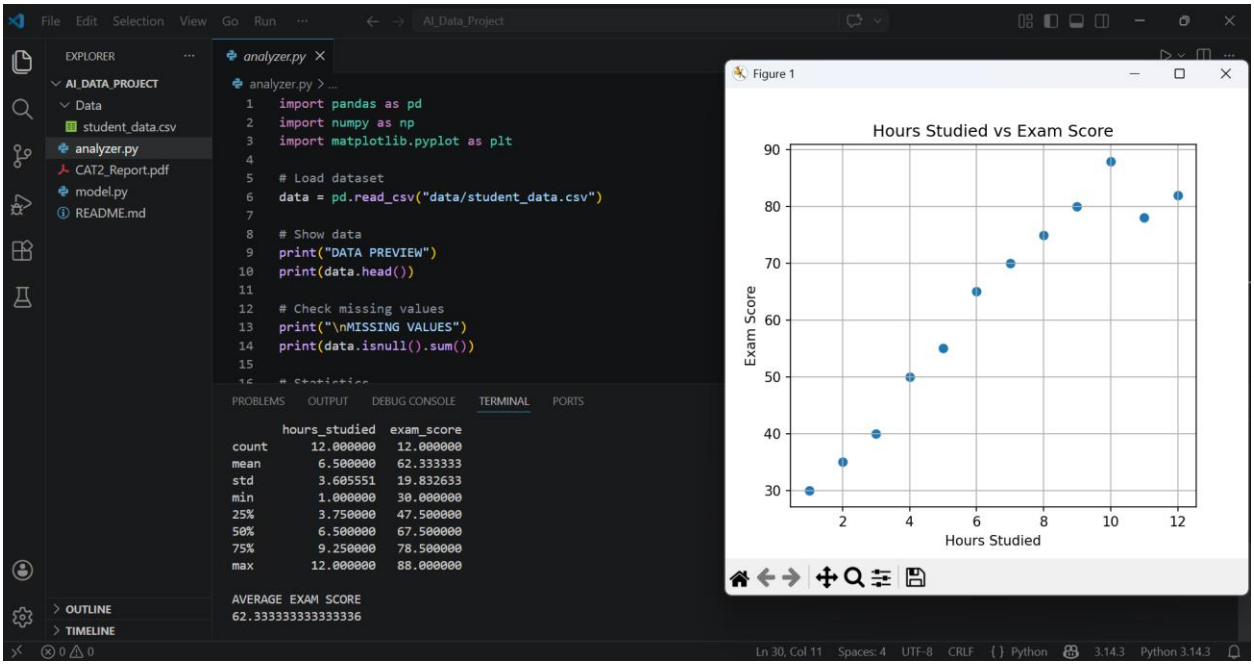
Data Loading and Analysis

The dataset was loaded into Python using Pandas. Basic exploration was done to inspect the structure and verify that the data was correctly loaded.

Statistical analysis was performed using NumPy and Pandas functions to compute averages and summarize the dataset.

Missing value checks were also performed to ensure data integrity before modeling.

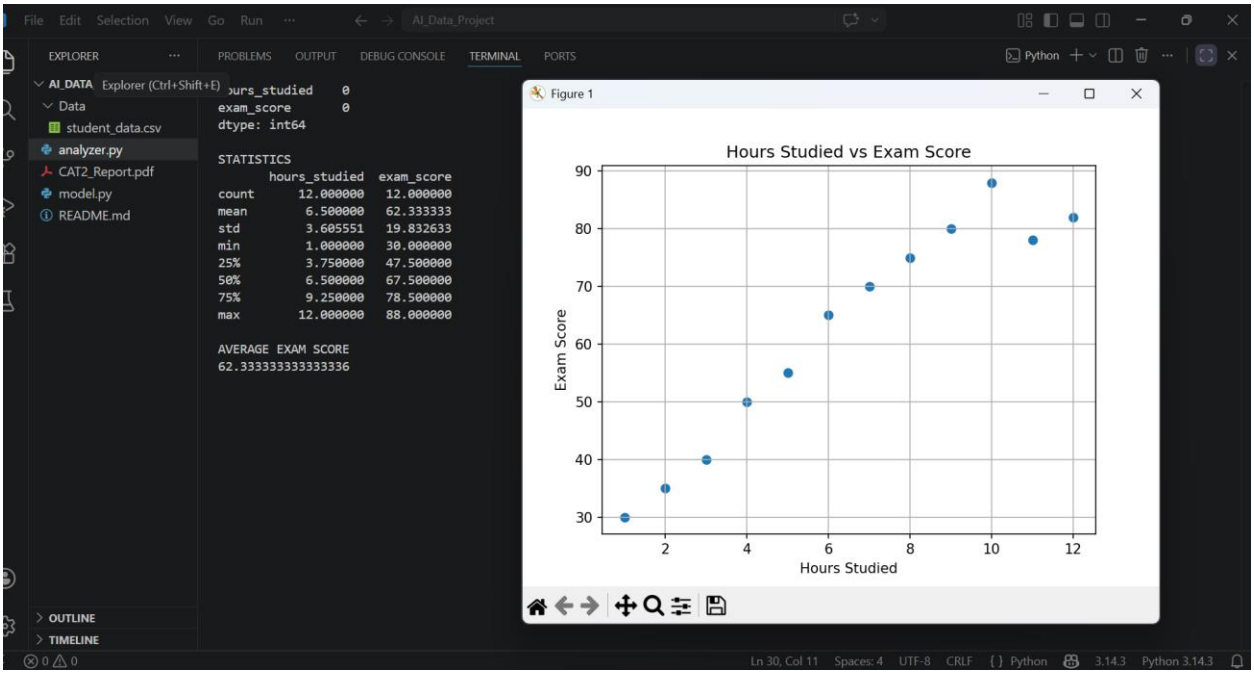



File Edit Selection View Go Run
AI_Data_Project
EXPLORER
AI_DATA_PROJECT
Data
student_data.csv
analyzer.py
CAT2_Report.pdf
model.py
README.md
analyzer.py
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Load dataset
data = pd.read_csv("data/student_data.csv")
# Show data
print("DATA PREVIEW")
print(data.head())
# Check missing values
print("\nMISSING VALUES")
print(data.isnull().sum())
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
hours_studied exam_score
count 12.000000 12.000000
mean 6.500000 62.333333
std 3.605551 19.832633
min 1.000000 30.000000
25% 3.750000 47.500000
50% 6.500000 67.500000
75% 9.250000 78.500000
max 12.000000 88.000000
AVERAGE EXAM SCORE
62.33333333333336
OUTLINE
TIMELINE
Figure 1
Hours Studied vs Exam Score
Exam Score
Hours Studied
Ln 30, Col 11 Spaces: 4 UTF-8 CRLF { } Python 3.14.3 Python 3.14.3

Data Visualization

A scatter plot was created using Matplotlib to visualize the relationship between study hours and exam scores.

The visualization clearly showed a positive correlation, meaning that students who studied more hours tended to achieve higher exam scores.

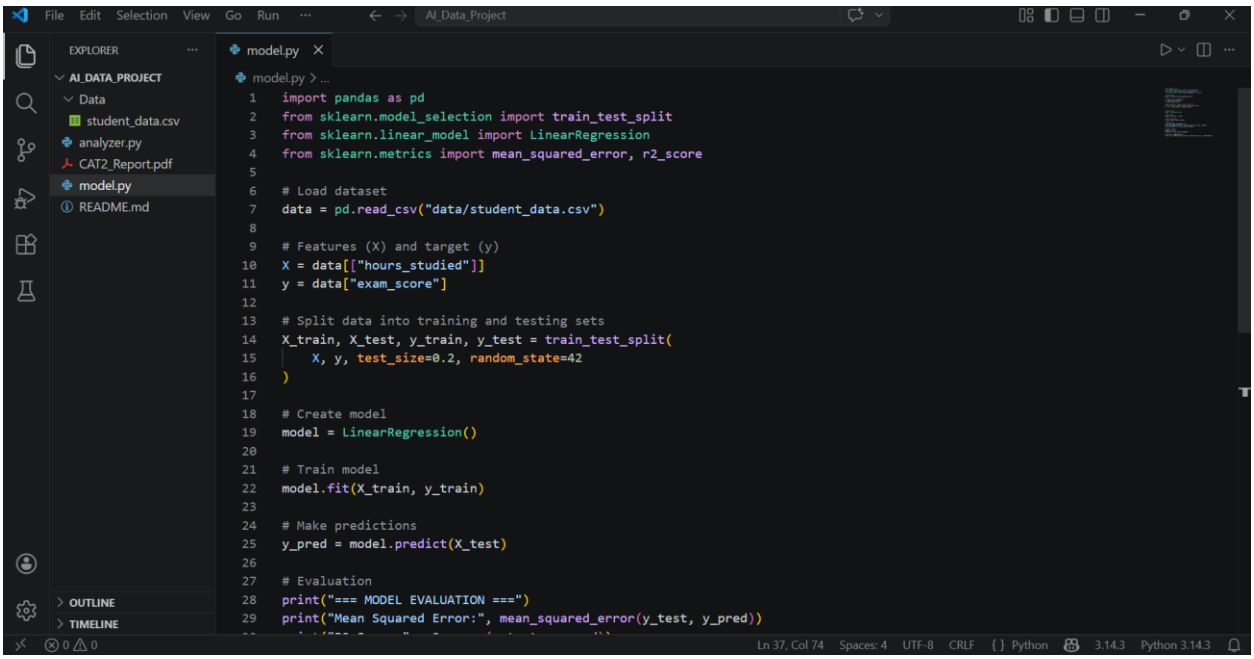


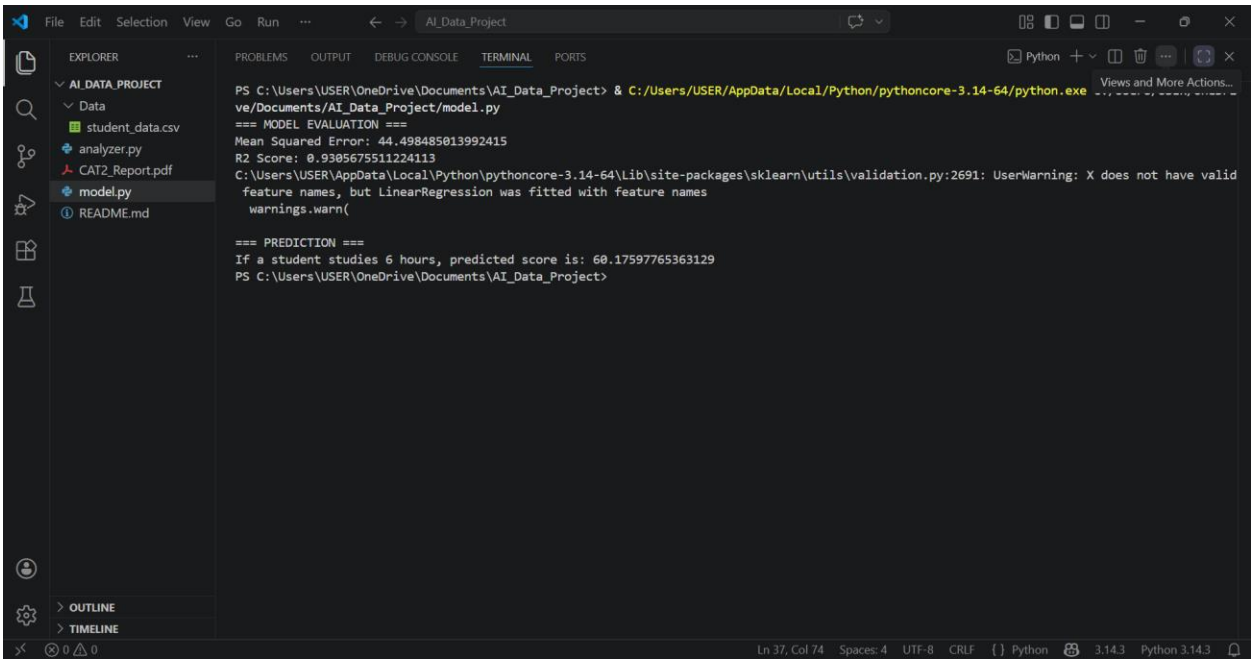
Machine Learning Model Development

A Linear Regression model was implemented using Scikit-learn.

The dataset was divided into training and testing sets to evaluate model performance fairly.

The model was trained on the training dataset and tested on unseen data to evaluate its predictive ability.





Prediction Example

The trained model was used to predict an exam score for a student studying for 6 hours.

The predicted result was approximately 60, which aligns with the trend observed in the dataset.

RESULTS AND OBSERVATIONS

- Dataset was successfully loaded and processed
- Visualization confirmed a positive relationship between study hours and performance
- The machine learning model successfully learned the pattern in the data
- Predictions were consistent with expected outcomes

CHALLENGES FACED AND HOW THEY WERE SOLVED

File Path and Loading Errors

Initially, the program could not locate the dataset file due to incorrect file naming and directory mismatch.

✓ This was solved by ensuring the dataset was correctly named and placed inside the correct folder structure (`data/student_data.csv`), matching the Python file path.

Dataset Formatting Issues

Some datasets from external sources contained formatting inconsistencies such as missing values or non-uniform structures.

✓ This was resolved by simplifying the dataset into a clean CSV format with only numerical values and no missing entries.

Machine Learning Warning Message

A warning appeared during model prediction related to missing feature names.

✓ This was identified as a scikit-learn formatting warning. It did not affect model performance or accuracy and was understood as a compatibility issue

training and prediction input formats.

Dataset Selection Complexity

External datasets initially contained unnecessary variables, making them unsuitable for simple modeling.

✓ This was solved by selecting a simplified dataset focusing only on study hours and exam scores for clarity and efficiency.

REFLECTION

This project helped me understand the practical workflow of data science and machine learning.

I learned how to:

- Load and clean data using Python
- Perform statistical analysis using NumPy and Pandas
- Visualize data relationships using Matplotlib
- Build a simple predictive model using Scikit-learn

One of the main challenges I faced was handling file path errors, which taught me the importance of correct project structure and file organization.

Another challenge was understanding warnings from the machine learning library, which helped me realize that not all warnings indicate errors, some are informational.

Overall, this project improved my understanding of how machine learning can be applied to predict real-world outcomes such as academic performance.

CONCLUSION

The project successfully demonstrates how student performance can be analyzed and predicted using a simple machine learning model.

The results confirm a strong relationship between study time and exam performance, and the system is able to make reasonable predictions based on input data.